Ziheng Duan

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EDUCATION

Ph.D. Computer Science, University of California, Irvine, 2021 – 2025

M.S. Computer Science, University of California, Irvine, 2021 – 2024

B.S. Automation & Computer Science (Dual Degree), Zhejiang University, 2016 – 2020

RESEARCH INTERESTS

Virtual Cell Modeling: Predicting cellular states and responses across diverse biological and patient contexts to support precision medicine.

Therapeutic Intervention Design: Identifying therapeutic targets and designing interventions to steer disease-associated cells toward healthier states.

Closed-Loop Scientific Discovery: Connecting hypothesis generation, experimental design, and model updates into a loop that learns from each round of experiments.

Methods: Graph learning, multimodal modeling, and large language models for single-cell and spatial omics.

WORK EXPERIENCE

Postdoctoral Researcher Starting Nov. 2026
Yale University, New Haven, CT
Lab of Prof. Mark Gerstein

Postdoctoral Researcher Mar. 2026 – Oct. 2026
University of California, Irvine, CA
Lab of Prof. Jing Zhang

- Developed TargetSage, an LLM reasoning agent for genome-scale therapeutic target identification that combines agentic evidence retrieval, GRPO-trained gene-level reasoning, and positive-unlabeled learning to correct false negatives among unlabeled druggable genes; outperforms ten baselines across 15 benchmark tasks covering 19,032 human protein-coding genes and prospectively ranks targets that later reached the clinic.
- Developed CellReprogrammer, a differentiable framework that optimizes sparse gene-perturbation sets to shift disease-associated cell states toward health, coupling a disease-state classifier with a foundation-model perturbation surrogate and relaxed top-K selection while co-optimizing druggability and on-target safety; evaluated across five psychiatric disorders and three brain cell types.

Consultant Apr. 2026 – Present
Genesis Molecular AI, San Diego, CA

- Built an autoimmune single-cell reference atlas spanning blood, gut, and skin across four studies (over 8M cells, 1,200+ donors) to profile target and pathway expression by immune cell type and disease, pairing donor-level statistical validation with eQTL–GWAS colocalization to support target evaluation.

Machine Learning Research Intern Jun. 2025 – Sep. 2025
Genesis Molecular AI, San Diego, CA

- Developed KNOT: A Knowledge Graph and Omics Integration Framework for Target Identification, integrating multi-omics data and biomedical knowledge graphs to enhance drug target prediction.

Graduate Researcher Sep. 2021 – Jun. 2025
University of California, Irvine, CA
Lab of Prof. Jing Zhang
Developed trustworthy computational methods to understand single-cell gene regulation, especially in spatial contexts, and their contributions to various diseases.

PUBLICATIONS

First Author Publications

[11] **Duan, Z.**, Wu, T., Li, X., Sun, S. D., Wang, C., Liu, J., Min, M. R., & Zhang, J. (2026). TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning. *NeurIPS-2026: Advances in Neural Information Processing Systems*.

[10] **Duan, Z.**, Li, X., Xiao, Z., Ying, R., & Zhang, J. (2025). MUSE: A Multi-slice Joint Analysis Method for Spatial Transcriptomics Experiments. *CIKM-2025: The 34th ACM International Conference on Information and Knowledge Management*. doi:10.1145/3746252.3761240

[9] **Duan, Z.**, Li, X., Zhang, Z., Song, J., & Zhang, J. (2025). DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion. *ICIP-2025: 2025 IEEE International Conference on Image Processing*. doi:10.1109/ICIP55913.2025.11084277

[8] **Duan, Z.**, Xu, S., Lee, C. Y., Riffle, D., & Zhang, J. (2024). iMIRACLE: An Iterative Multi-View Graph Neural Network to Model Intercellular Gene Regulation From Spatial Transcriptomic Data. *CIKM-2024: The 33rd ACM International Conference on Information and Knowledge Management*. doi:10.1145/3627673.3679574

[7] **Duan, Z.**, Riffle, D., Li, R., Liu, J., Min, M. R., & Zhang, J. (2024). Impeller: a path-based heterogeneous graph learning method for spatial transcriptomic data imputation. *Bioinformatics*, btae339. doi:10.1093/bioinformatics/btae339

[6] **Duan, Z.**, Xu, S., Sai Srinivasan, S., Hwang, A., Lee, C. Y., Yue, F., ... & Zhang, J. (2024). scENCORE: leveraging single-cell epigenetic data to predict chromatin conformation using graph embedding. *Briefings in Bioinformatics*, 25(2), bbae096. doi:10.1093/bib/bbae096

[5] **Duan, Z.**, Lee, C. Y., & Zhang, J. (2023). ExAD-GNN: Explainable Graph Neural Network for Alzheimer's Disease State Prediction from Single-cell Data. *APSIPA Transactions on Signal and Information Processing*, 12(5). doi:10.1561/116.00000239

[4] **Duan, Z.**, Dai, Y., Hwang, A., Lee, C. Y., Xie, K., Xiao, C., ... & Zhang, J. (2023). iHerd: an integrative hierarchical graph representation learning framework to quantify network changes and prioritize risk genes in disease. *PLOS Computational Biology*, 19(9), e1011444. doi:10.1371/journal.pcbi.1011444

[3] **Duan, Z.**, Xu, H., Huang, Y., Feng, J., & Wang, Y. (2023). Multivariate Time Series Forecasting with Transfer Entropy Graph. *Tsinghua Science and Technology*, 28(1), 141–149. doi:10.26599/TST.2021.9010081

[2] **Duan, Z.**, Xu, H., Wang, Y., Huang, Y., Ren, A., Xu, Z., ... & Wang, W. (2022). Multivariate time-series classification with hierarchical variational graph pooling. *Neural Networks*, 154, 481–490. doi:10.1016/j.neunet.2022.07.032

[1] **Duan, Z.**, Wang, Y., Ye, W., Fan, Q., & Li, X. (2022). Connecting latent relationships over heterogeneous attributed network for recommendation. *Applied Intelligence*, 52(14), 16214–16232. doi:10.1007/s10489-022-03340-7

Other Publications

[24] Li, X., Hu, Y., **Duan, Z.**, Wang, X., Sun, S. D., Sikdar, D., Liu, Y., & Zhang, J. (2026). sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics. *NeurIPS-2026: Advances in Neural Information Processing Systems*. openreview.net/forum?id=Hk85ZGSGbQ

[23] Lee, C. Y., Hwang, A., ..., **Duan, Z.**, ..., & Girgenti, M. J. (2026). Central amygdala single-nucleus atlas reveals chromatin and gene transcription dynamics in human alcohol use disorder. *Nature Communications*. doi:10.1038/s41467-026-68351-1

[22] Zou, Z., & **Duan, Z.** (2025). SpaLLM: a general framework for spatial domain identification with large language models. *Frontiers in Bioinformatics*, 5, 1713975. doi:10.3389/fbinf.2025.1713975

[21] Gaynor-Gillett, S. C., ..., **PsychENCODE Consortium**, ... & White, K. P. (2025). A map of enhancer regions in primary human neural progenitor cells using capture STARR-seq. *Genome Research*, 35(8), 1887–1901. doi:10.1101/gr.279584.124

[20] Hwang, A., Skarica, M., ..., **Duan, Z.**, ..., Zhang, J., & Girgenti, M. J. (2025). Single-cell transcriptomic and chromatin dynamics of the human brain in PTSD. *Nature*. doi:10.1038/s41586-025-09083-y

[19] Liu, J., Xu, S., Riffle, D., **Duan, Z.**, Min, M. R., & Zhang, J. (2024). Understanding Transcriptional Regulatory Redundancy by Learnable Global Subset Perturbations. *ACML-2024: Proceedings of the 16th Asian Conference on Machine Learning*, PMLR 260:383–398. openreview.net/forum?id=2ODIoB7Ijg

[18] Li, X., **Duan, Z.**, Dai, Y., Xu, S., & Zhang, J. (2024). iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics. *BMVC-2024: The 35th British Machine Vision Conference*. bmva-archive.org.uk/bmvc/2024/papers/Paper_991

[17] Dai, Y., **Duan, Z.**, Xu, S., & Zhang, J. (2024). CytoFlow: A Novel Computational Method to Construct Signal Transduction Networks at Single-Cell Resolution Based on Flow Networks. *MedAI-2024: IEEE International Conference on Medical Artificial Intelligence*, 212–217. doi:10.1109/MedAI62885.2024.00034

[16] Wen, C., Margolis, M., Dai, R., Zhang, P., Przytycki, P. F., Vo, D. D., ..., **PsychENCODE Consortium**, ... & Gandal, M. J. (2024). Cross-ancestry atlas of gene, isoform, and splicing regulation in the developing human brain. *Science*, 384(6698), eadh0829. doi:10.1126/science.adh0829

[15] Deng, C., Whalen, S., Steyert, M., Ziffra, R., Przytycki, P. F., Inoue, F., ..., **PsychENCODE Consortium**, ... & Pollard, K. S. (2024). Massively parallel characterization of regulatory elements in the developing human cortex. *Science*, 384(6698), eadh0559. doi:10.1126/science.adh0559

[14] Huuki-Myers, L. A., Spangler, A., Eagles, N. J., Montgomery, K. D., Kwon, S. H., Guo, B., ..., **PsychENCODE Consortium**, ... & Maynard, K. R. (2024). A data-driven single-cell and spatial transcriptomic map of the human prefrontal cortex. *Science*, 384(6698), eadh1938. doi:10.1126/science.adh1938

[13] Ruzicka, W. B., Mohammadi, S., Fullard, J. F., Davila-Velderrain, J., Subburaju, S., Tso, D. R., ..., **PsychENCODE Consortium**, ... & Kellis, M. (2024). Single-cell multi-cohort dissection of the schizophrenia transcriptome. *Science*, 384(6698), eadg5136. doi:10.1126/science.adg5136

[12] Emani, P. S., Liu, J. J., Clarke, D., Jensen, M., Warrell, J., Gupta, C., ..., **Duan, Z.**, ... & Gerstein, M. (2024). Single-cell genomics and regulatory networks for 388 human brains. *Science*, 384(6698), eadi5199. doi:10.1126/science.adi5199

[11] Dai, R., Chu, T., Zhang, M., Wang, X., Jourdon, A., Wu, F., ..., **PsychENCODE Consortium**, ... & Liu, C. (2024). Evaluating performance and applications of sample-wise cell deconvolution methods on human brain transcriptomic data. *Science Advances*, 10(21), eadh2588. doi:10.1126/sciadv.adh2588

[10] Pratt, H. E., Andrews, G., Shedd, N., Phalke, N., Li, T., Pampari, A., ..., **PsychENCODE Consortium**, ... & Weng, Z. (2024). Using a comprehensive atlas and predictive models to reveal the complexity and evolution of brain-active regulatory elements. *Science Advances*, 10(21), eadj4452. doi:10.1126/sciadv.adj4452

[9] Xia, Y., Xia, C., Jiang, Y., Chen, Y., Zhou, J., Dai, R., ..., **PsychENCODE Consortium**, ... & Chen, C. (2024). Transcriptomic sex differences in postmortem brain samples from patients with psychiatric disorders. *Science Translational Medicine*, 16(749), eadh9974. doi:10.1126/scitranslmed.adh9974

[8] Lee, C. Y., Riffle, D., Xiong, Y., Momtaz, N., Lei, Y., Pariser, J. M., ..., **Duan, Z.**, ... & Zhang, J. (2024). Characterizing dysregulations via cell-cell communications in Alzheimer's brains using single-cell transcriptomes. *BMC Neuroscience*, 25(1), 24. doi:10.1186/s12868-024-00867-y

[7] Zhang, R., **Duan, Z.**, Lee, C. Y., Riffle, D., Min, M. R., & Zhang, J. (2023). Turtling: a time-aware neural topic model on NIH grant data. *Bioinformatics Advances*, vbad096. doi:10.1093/bioadv/vbad096

[6] Lee, C. Y., Chen, Y., **Duan, Z.**, Xu, M., Girgenti, M. J., Xu, K., ... & Zhang, J. (2022). Venus: An efficient virus infection detection and fusion site discovery method using single-cell and bulk RNA-seq data. *PLOS Computational Biology*, 18(10), e1010636. doi:10.1371/journal.pcbi.1010636

[5] Wang, Y., **Duan, Z.**, Huang, Y., Xu, H., Feng, J., & Ren, A. (2022). MTHetGNN: A heterogeneous graph embedding framework for multivariate time series forecasting. *Pattern Recognition Letters*, 153, 151–158. doi:10.1016/j.patrec.2021.12.008

[4] Xu, H., **Duan, Z.**, Wang, Y., Feng, J., Chen, R., Zhang, Q., & Xu, Z. (2021). Graph partitioning and graph neural network based hierarchical graph matching for graph similarity computation. *Neurocomputing*, 439, 348–362. doi:10.1016/j.neucom.2021.01.068

[3] Liao, Z., Yang, Y., Sun, C., Wu, R., **Duan, Z.**, Wang, Y., ... & Xu, J. (2021). Image-based prediction of granular flow behaviors in a wedge-shaped hopper by combing DEM and deep learning methods. *Powder Technology*, 383, 159–166. doi:10.1016/j.powtec.2021.01.041

[2] Wang, B., Zhou, C., **Duan, Z.**, Zhu, Q., Wu, J., & Xiong, R. (2020). Untethered quadrupedal hopping and bounding on a trampoline. *Frontiers of Mechanical Engineering*, 15, 181–192. doi:10.1007/s11465-019-0559-5

[1] Wang, Y., **Duan, Z.**, Liao, B., Wu, F., & Zhuang, Y. (2019). Heterogeneous Attributed Network Embedding with Graph Convolutional Networks. *AAAI-2019: Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01), 10061–10062. doi:10.1609/aaai.v33i01.33010061

TALKS

- "Decoding Disease, Designing Interventions."
  ICS Honors Seminar, University of California, Irvine — Oct. 2026
- "From Immune Atlases to Therapeutic Reprogramming."
  Genesis Molecular AI (internal talk) — Sep. 2026
- "AI-Driven Therapeutic Target Discovery and Cell-State Reprogramming."
  SCORCH 2026 Fall Meeting, San Diego, CA — Sep. 2026
- "scENCORE: Leveraging Single-Cell Epigenetic Data to Predict Chromatin Conformation Using Graph Embedding."
  13th Annual Symposium, Southern California Systems Biology (invited) — Mar. 2024
- "iHerd: An Integrative Hierarchical Graph Representation Learning Framework to Quantify Network Changes and Prioritize Risk Genes in Disease."
  2023 SCORCH Consortium Meeting (invited) — Dec. 2023

AWARDS AND HONORS

- Travel Grant, IEEE International Conference on Image Processing (ICIP 2025) — 2025
- Dean's Fellowship, University of California, Irvine — 2021
- Research and Innovation Scholarship, Zhejiang University — 2019
- Academic Excellence Scholarship, Zhejiang University — 2017 – 2019

TEACHING

- *Reader*, University of California, Irvine — Spring 2022
  ICS 6D: Discrete Mathematics for Computer Science
- *Reader*, University of California, Irvine — Winter 2022
  ICS 178: Machine Learning and Data Mining
- *Reader*, University of California, Irvine — Fall 2021
  ICS 178: Machine Learning and Data Mining

MENTORING

- *Mentor*, Google Summer of Code 2026 — Summer 2026
  StaR: A Stability-Aware Representation Learning Framework for Spatial Domain Identification
  Guided the development of StaR, a plug-in training framework for improving the stability and reproducibility of spatial domain identification across random initializations.

- *Mentor*, Google Summer of Code 2025 — Summer 2025
  RAG-ST: Retrieval-Augmented Generation for Spatial Transcriptomics
  Guided the development of a retrieval-augmented framework for predicting spatial gene expression from histology images, focusing on interpretability and modular design.

- *Mentor*, Google Summer of Code 2024 — Summer 2024
  BenchmarkST: Spatial Transcriptomics Gene Imputation Benchmarking
  Supervised the creation of a benchmarking suite for evaluating cross-platform, multi-species gene expression imputation methods.

SERVICES

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